

SMART INDIA HACKATHON 2026

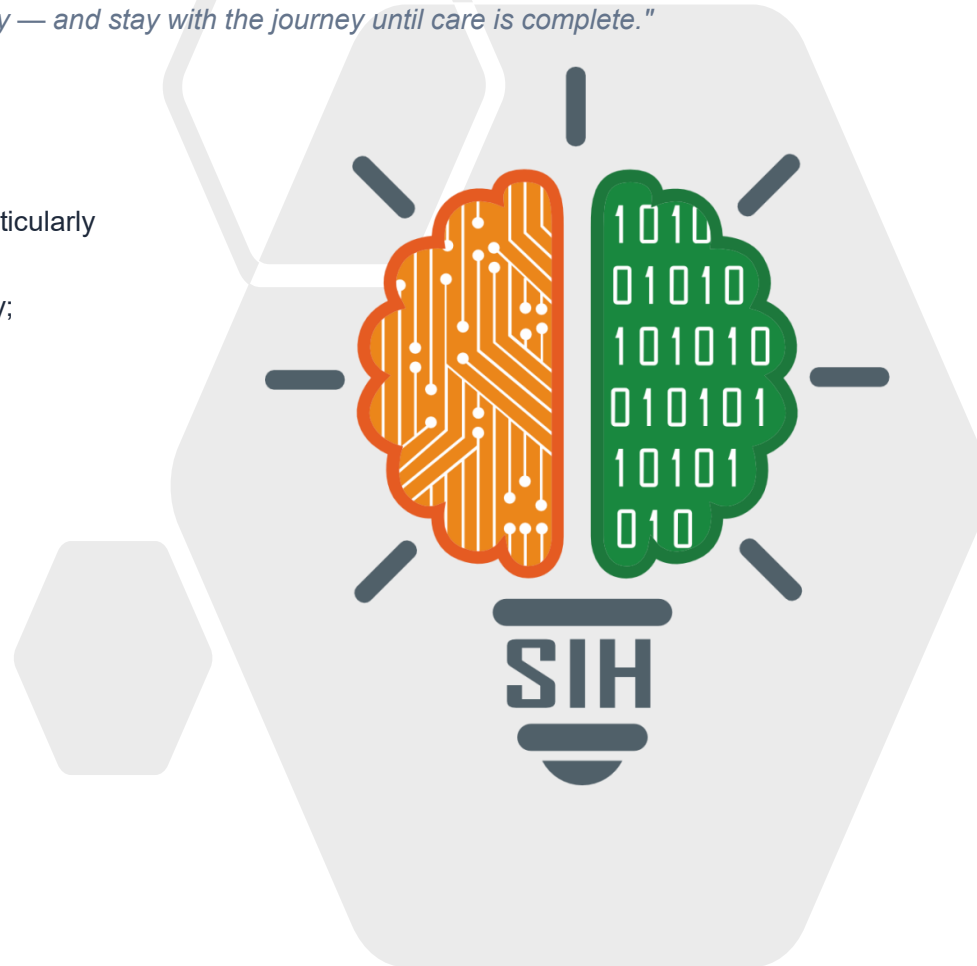
SWASTHYASETU (स्वास्थ्यसेतु)



Rural Healthcare Referral & Care Continuity Platform

"Don't just send the patient. Make sure the care is ready — and stay with the journey until care is complete."

- **Problem Statement ID:** SIH26133
- **Problem Statement Title:** Accessibility and quality of public healthcare services, particularly in rural and underserved areas
- **Nodal Authority:** Government of Maharashtra (Maharashtra State Innovation Society; Department of Skills, Employment, Entrepreneurship and Innovation)
- **Theme:** MedTech / BioTech / HealthTech
- **PS Category:** Software
- **Team ID:** SIH2026-TEAM-2
- **Team Name:** SwasthyaSetu (Team 2)



Proposed Solution (Describe your Idea/Solution/Prototype) — Detailed Explanation | Problem Resolution | Uniqueness & Innovation

THE PROBLEM: The Rural Healthcare 'Clinical Black Hole'

- 150,000+ blind paper referrals weekly across India's 31,000 PHCs without receiving notice.
- 48.2% Referral Drop-out: Patients travel 40-80 km only to find specialists absent or beds full.
- Zero Counter-Referral: Discharges never reach village ASHAs, causing fatal relapses & OOPE debt.

THE SOLUTION: AI-Powered Closed-Loop Care Orchestration

- End-to-End Care Continuity: Connects Sub-Centre (AAM) -> PHC -> CHC -> District Hospital.
- Care Readiness Verification: Verifies 6-factor readiness before travel + auto-reroutes if blocked.
- Zero-Typing & Zero-Data-Loss: Multilingual voice intake in Marathi + offline-first sync queue.

Pillar 1: Voice-First Vernacular

Tech: Bhashini Indic STT / NMT

Action: Spoken Marathi/Hindi symptom dictation (e.g. 'छातीत दुखणे, धाप लागणे').

Impact: Eliminates complex English text entry; reduces ASHA logging from 8m to 50s on basic Android Go phones.

Pillar 2: Patient Offline-First

Tech: SQLite / WatermelonDB CRDTs

Action: Local encrypted queue storing 5,000+ patient records on mobile device.

Impact: Guaranteed zero data loss in rural monsoon network dead zones; automatic background sync on reconnect.

Pillar 3: Coordinated One-Trip

Tech: Care Bundle Decision Model

Action: Bundles specialty consultation, on-site diagnostics (ECG, CBC), and documents.

Impact: Stops the 'ping-pong' referral cycle; cuts travel & catastrophic out-of-pocket costs by >60%.

Pillar 4: Care Readiness (WOW)

Tech: 6-Factor Decision Engine

Action: Checks Doctor, Diagnostics, Equipment, ICU Beds (<85%), and Docs before departure.

Impact: Stops door refusals; dynamic rerouting recovers referral if hospital becomes unavailable.

KEY NOVELTY: The ONLY system providing Closed-Loop Tracking (Village -> Hospital -> ASHA Counter-Referral) + ABDM UHI & MJPJAY Integration



Technologies Used (Languages, Frameworks, Architecture) & Methodology Process (Flowcharts / Working Prototype Pipeline)



Architecture Layer	Production Technology Stack	Engineering & Operational Rationale
Mobile & Web Client	Vite + React 18 PWA / React Native Android Go	Ultra-lightweight (<15 MB APK), offline-ready, runs smoothly on 2GB RAM phones.
API Gateway & Services	NestJS (Node.js 20+) Modular Microservices Architecture	High-throughput asynchronous REST/GraphQL gateway, JWT + Argon2 security, DLT SMS.
Persistence & GIS	PostgreSQL + PostGIS (Spatial) + Prisma ORM	ACID transactions, spatial hospital distance queries, drive-time and load balancing.
Offline Storage & Queue	SQLite / WatermelonDB (CRDTs) + Redis BullMQ	Guaranteed local storage during network drops; event-based background synchronization.
AI & Intelligence Layer	Gemini 2.5 Flash + Bhashini Indic STT + Tesseract OCR	Sub-second symptom extraction, vernacular Marathi voice dictation, scanned slip OCR.
Protocols & Security	ABDM Gateway (ABHA, HFR, HPR), FHIR R4, DPDP Act 2023	National Health Authority compliant, encrypted health bundles, verifiable patient consent.

Analysis of Idea Feasibility, Potential Challenges & Risks, and Strategic Mitigation Measures

1. Operational Feasibility (Frontline Ready)

- Replaces 8 duplicate paper registers with 50-second Marathi voice dictation.
- Operates on existing government smartphones without purchasing specialized tablets.
- Designed for low-literacy ASHAs with audio cues, icons, and automated SMS.

2. Technical Feasibility (Interoperable Rails)

- Integrates directly with live national APIs: ABDM (67 Cr ABHA IDs) & Bhashini.
- CRDT local replication guarantees zero data corruption during partial syncs.
- Runs on proven open-source web and mobile architecture with MeitY cloud compliance.

3. Financial Viability (B2G Unit Economics)

- Funded via State NHM PIP Innovation budget + ABDM DHIS incentives (₹500/bed/mo).
- District Operating Cost: ~₹18 Lakh/year vs. State Contract Value: ~₹35 Lakh/year.
- 48.5% operating margin supports field engineers and frontline worker training.

Risk Category	Identified Challenge	Strategic Mitigation Plan
Technical Risk	Cellular blackout >7 days in deep tribal monsoon areas.	Local SQLite retains 5,000 records encrypted; offline QR token + peer-to-peer sync at weekly PHC meet.
Operational Risk	Hospital specialist ignores incoming referral queue.	Auto-escalation alert to Civil Surgeon dashboard after 4 hours + SMS reminder + physical priority token.
Clinical Risk	Triage rule misclassifies acute case as low risk.	Deterministic ICMR/WHO CDSS thresholds; mandatory human-in-the-loop doctor authorization; no autonomous diagnosis.
Adoption Risk	Low smartphone literacy among senior village ASHAs.	Marathi voice dictation + visual iconographic UI + ₹50 performance incentive per verified referral follow-up.

Potential Impact on Target Audience & Comprehensive Benefits (Social, Economic, Environmental, Administrative)

< 10%
Referral Drop-out

Slashed from 48.2% baseline via closed-loop tracking & active navigation.

60%+
OOPE Travel Reduction

Saves rural families from debt traps via Coordinated One-Trip Care Bundles.

< 45 Min
Hospital Wait Time

Fast-track casualty triage bypasses 3.5-hour general OPD queues.

100%
Follow-up Compliance

Automated counter-referral tasks sent to village ASHAs on patient discharge.

SOCIAL IMPACT: Maternal & Cardiac Lives Saved

- Prevents maternal mortality (PPH, eclampsia) & cardiac deaths in tribal blocks (Melghat, Nandurbar).
- Eliminates fear and confusion for illiterate citizens through vernacular voice guidance.
- Empowers frontline women (ASHAs) as digital healthcare leaders in their community.

ECONOMIC IMPACT: Eliminating Healthcare Debt

- Health expenses push 55 million Indians into poverty annually; platform cuts private jeep & repeat test costs.
- Reduces tertiary hospital over-utilization by triaging treatable cases at primary PHC/CHC level.
- Unlocks ABDM DHIS federal cash incentives (₹500/bed/mo) for participating state hospitals.

ENVIRONMENTAL IMPACT: Paperless Green Health

- Replaces millions of physical registers and paper referral slips with digital records.
- Significantly cuts vehicular emissions and fuel consumption from repeated, failed hospital trips.
- Eco-friendly server architecture optimized for low-power edge execution.

ADMINISTRATIVE IMPACT: Real-Time Governance

- Live District Health Officer (DHO) bottleneck heatmaps surface specialist shortages & equipment outages.
- Replaces delayed 45-day HMIS aggregate reports with actionable operational telemetry.
- Provides accountability across secondary facilities for referral turnaround and bed readiness.

Details & Links of Reference Research Work, Competitive Benchmarking Matrix, and Epidemiological Evidence

Key Capability Dimension	eSanjeevani (C-DAC)	NIC e-Hospital	Khushi Baby / CHT	108 EMS Fleet	SwasthyaSetu
Closed-Loop Physical Referral	✗ None (Tele-advice only)	✗ None (Internal OPD only)	⚠ Partial (MCH only)	✗ Transport only	✓ Full Closed-Loop Tracking
Offline-First Mobile Architecture	✗ Fails without 4G	✗ Desktop Intranet only	✓ Offline Sync	⚠ Voice Call based	✓ Offline SQLite + CRDT Sync
Vernacular Voice Triage	✗ None	✗ None	✗ None	✗ None	✓ Bhashini Marathi/Hindi STT
Pre-Arrival Casualty Alert	✗ None	✗ None	✗ None	⚠ Manual Phone Call	✓ Real-Time Triage Card
Automated ASHA Counter-Referral	✗ None	✗ None	✗ None	✗ None	✓ Discharge to ASHA Task

1. MoHFW Rural Health Statistics

Source: Health Dynamics of India 2022-23

Finding: Acute 79.9% shortfall of specialists at rural CHCs nationwide.

Relevance: Validates why capacity-aware routing is the only viable allocation mechanism.

2. The Lancet Global Health

Source: Referral Pathways in Rural India (2020)

Finding: 48.2% of referred primary patients never reach secondary hospitals.

Relevance: Provides empirical baseline for SwasthyaSetu closed-loop tracking.

3. National Health Accounts

Source: NHA 2020-21 / Lancet Public Health

Finding: OOPE accounts for 47.1% of health spend; pushes 55M into poverty.

Relevance: Proves need for Coordinated One-Trip care to slash private transport costs.

4. JMIR 2025 & ABDM Specs

Source: Li et al., JMIR 2025; ABDM FHIR R4

Finding: Bidirectional referral cuts transfer delay from 2.51 to 0.90 days.

Relevance: Core blueprint for SwasthyaSetu 3-way visibility & ABHA integration.